CCNA Practice Final 1

Implementación de redes de área amplia y servicios distribuidos (TC3003B.570)

Fernando Daniel Monroy Sánchez (A01750536)

22 de octubre de 2026

Profesor: Juan Manuel Campos Sandoval

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1. Which two statements describe the characteristics of fiber-optic cabling? (Choose two.)
   - **Respuesta: Fiber-optic cabling does not conduct electricity.; Fiber-optic cabling is primarily used as backbone cabling.**
   - El cable de fibra óptica utiliza hilos flexibles de vidrio o plástico transparente para transmitir señales luminosas en lugar de pulsos eléctricos.
   - Como el núcleo y el revestimiento están hechos de vidrio aislante, la fibra no conduce electricidad y resulta inmune a la interferencia electromagnética y de radiofrecuencia.
   - La fibra óptica se utiliza primordialmente como cableado dorsal o troncal para unir edificios, cuartos de telecomunicaciones y enlaces de larga distancia entre redes.
2. Although CSMA/CD is still a feature of Ethernet, why is it no longer necessary?
   - **Respuesta: the use of full-duplex capable Layer 2 switches**
   - Las redes Ethernet antiguas con concentradores (hubs) operan en modo semidúplex, donde solo 1 dispositivo puede enviar o recibir a la vez en el medio compartido.
   - Si 2 dispositivos transmiten al mismo tiempo en un medio compartido ocurre una colisión, por lo que las redes en semidúplex requieren CSMA/CD para detectar colisiones.
   - Los switches Ethernet de Capa 2 y las tarjetas NIC actuales usan comunicación en dúplex completo, donde los 2 extremos envían y reciben simultáneamente sin colisiones.
3. Which two acronyms represent the data link sublayers that Ethernet relies upon to operate? (Choose two.)
   - **Respuesta: MAC; LLC**
   - Los estándares IEEE 802 para redes LAN y MAN dividen la capa de enlace de datos del modelo OSI en 2 subcapas diferenciadas: LLC y MAC.
   - La subcapa de control de enlace lógico (LLC, IEEE 802.2) comunica el software de las capas superiores con el hardware de las capas inferiores e identifica el protocolo de red encapsulado.
   - La subcapa de control de acceso al medio (MAC, IEEE 802.3) se encarga de la encapsulación de datos, el direccionamiento físico MAC y el acceso a los medios físicos de transmisión.
4. A network team is comparing topologies for connecting on a shared media. Which physical topology is an example of a hybrid topology for a LAN?
   - **Respuesta: extended star**
   - En las redes LAN de acceso múltiple, los dispositivos finales se conectan a un dispositivo intermedio central como un switch Ethernet para formar una topología en estrella.
   - Una topología en estrella extendida amplía la topología en estrella porque interconecta múltiples switches Ethernet en un diseño físico híbrido.
   - Las topologías de bus encadenan los sistemas finales entre sí en 1 cable, y las topologías de anillo conectan cada sistema con su vecino formando un anillo.
5. What does a router do when it receives a Layer 2 frame over the network medium?
   - **Respuesta: de-encapsulates the frame**
   - Cuando un router acepta una trama desde un medio físico, de inmediato desencapsula el encabezado y el tráiler de Capa 2 para extraer la PDU de Capa 3.
   - Después, el router examina la dirección IP de destino del paquete y busca en su tabla de enrutamiento

la entrada con la mejor coincidencia.
- Por último, el router reencapsula el paquete dentro de una nueva trama y reenvía esa trama hacia el siguiente segmento de la red física.

6. What routing table entry has a next hop address associated with a destination network?
   - **Respuesta: remote routes**
   - La tabla de enrutamiento de un router almacena 3 tipos de entradas de ruta: redes conectadas directamente, redes remotas y una ruta predeterminada.
   - Las entradas de redes conectadas directamente representan interfaces activas del propio router local, por lo que no requieren una dirección de router de siguiente salto.
   - Las entradas de redes remotas representan redes de destino conectadas a otros routers, por lo que el router reenvía los paquetes a la dirección IP del siguiente salto.
7. Which term describes a field in the IPv4 packet header that contains a unicast, multicast, or broadcast address?
   - **Respuesta: destination IPv4 address**
   - El encabezado del paquete IPv4 de 20 bytes contiene campos binarios que el proceso de Capa 3 examina para entregar el paquete al dispositivo final de destino.
   - El campo de dirección IPv4 de destino contiene un valor binario de 32 bits que representa la dirección unicast, multicast o broadcast del receptor.
   - En cambio, el campo de dirección IPv4 de origen identifica siempre a 1 host emisor, mientras que TTL, Protocolo y Suma de comprobación almacenan valores de control.
8. Refer to the exhibit. PC1 issues an ARP request because it needs to send a packet to PC3. In this scenario, what will happen next?
   - **Respuesta: RT1 will send an ARP reply with its own Fa0/0 MAC address.**
   - Cuando la dirección IPv4 de destino (PC3) pertenece a una red distinta a la de origen (PC1), PC1 debe enviar la trama a su puerta de enlace predeterminada (RT1).
   - Si la entrada de la puerta de enlace predeterminada no figura en la tabla ARP, PC1 envía una solicitud ARP para obtener la dirección MAC de la puerta de enlace.
   - Solo el dispositivo con la dirección IPv4 solicitada responde, por lo que el router RT1 envía una respuesta ARP unicast con su propia dirección MAC de la interfaz Fa0/0.
9. Refer to the exhibit. The exhibit shows a small switched network and the contents of the MAC address table of the switch. PC1 has sent a frame addressed to PC3. What will the switch do with the frame?
   - **Respuesta: The switch will forward the frame to all ports except port 4.**
   - Un switch Ethernet de Capa 2 reenvía tramas buscando una coincidencia entre la dirección MAC de destino de la trama y una entrada en su tabla de direcciones MAC.
   - Si la dirección MAC unicast de destino no está en la tabla, el switch reenvía la trama por todos los puertos excepto por el puerto de entrada.
   - Como la tabla MAC no contiene la dirección MAC de PC3, el switch inunda esta trama unicast desconocida por todos los puertos excepto el puerto 4 de entrada.
10. Match the command with the device mode at which the command is entered.
    - **Respuesta: service password-encryption -> R1(config)#; enable -> R1>; copy running-config startup-config -> R1#; login -> R1(config-line)#; ip address 192.168.4.4 255.255.255.0 -> R1(config-if)#**
    - El modo EXEC usuario `R1>` ofrece comandos de diagnóstico básicos, donde `enable` permite pasar al modo EXEC privilegiado `R1#`.
    - En el modo EXEC privilegiado `R1#` se gestionan archivos de arranque y ejecución mediante comandos como `copy running-config startup-config`.
    - El modo de configuración global `R1(config)#` aplica parámetros generales al equipo como `service password-encryption`.
    - Los modos de subconfiguración específicos aplican directivas concretas: `R1(config-line)#` para habilitar `login` en líneas de terminal y consola, y `R1(config-if)#` para asignar la dirección IP y máscara a una interfaz con `ip address`.
11. Refer to the exhibit. A network administrator is connecting a new host to the Registrar LAN. The host needs to communicate with remote networks. What IP address would be configured as the default gateway on the new host?
    - **Respuesta: 192.168.235.234**

- Un host necesita una dirección de puerta de enlace predeterminada para alcanzar redes remotas, y esa dirección corresponde a la interfaz del router conectada a la red local.
- En un router Cisco, el comando `description` identifica la red conectada y el comando `ip address` asigna la dirección IPv4 y máscara a esa interfaz.
- La interfaz `gi0/1` tiene la descripción `Connects to the Registrar LAN` y la dirección IPv4 `192.168.235.234`, por lo que los hosts en esa LAN deben usar `192.168.235.234`.

Anexo: Capturas del material de Cisco Introduction to Networks

7.4.4 Duplex and Speed Settings

Two of the most basic settings on a switch are the bandwidth (sometimes referred to as "speed") and duplex settings for each individual switch port. It is critical that the duplex and bandwidth settings match between the switch port and the connected devices, such as a computer or another switch.

There are two types of duplex settings used for communications on an Ethernet network:

- **Full-duplex** - Both ends of the connection can send and receive simultaneously.
- **Half-duplex** - Only one end of the connection can send at a time.

Autonegotiation is an optional function found on most Ethernet switches and NICs. It enables two devices to automatically negotiate the best speed and duplex capabilities. Full-duplex is chosen if both devices have the capability along with their highest common bandwidth.

In the figure, the Ethernet NIC for PC-A can operate in full-duplex or half-duplex, and in 10 Mbps or 100 Mbps.



6.2.4 LAN Topologies

In multiaccess LANs, end devices (i.e., nodes) are interconnected using star or extended star topologies, as shown in the figure. In this type of topology, end devices are connected to a central intermediary device, in this case, an Ethernet switch. An **extended star** extends this topology by interconnecting multiple Ethernet switches. The star and extended topologies are easy to install, very scalable (easy to add and remove end devices), and easy to troubleshoot. Early star topologies interconnected end devices using Ethernet hubs.

At times there may be only two devices connected on the Ethernet LAN. An example is two interconnected routers. This would be an example of Ethernet used on a point-to-point topology.

Legacy LAN Topologies

Early Ethernet and legacy Token Ring LAN technologies included two other types of topologies:

- **Bus** - All end systems are chained to each other and terminated in some form on each end. Infrastructure devices such as switches are not required to interconnect the end devices. Legacy Ethernet networks were often bus topologies using coax cables because it was inexpensive and easy to set up.
- **Ring** - End systems are connected to their respective neighbor forming a ring. The ring does not need to be terminated, unlike in the bus topology. Legacy Fiber Distributed Data Interface (FDDI) and Token Ring networks used ring topologies.

The figures illustrate how end devices are interconnected on LANs. It is common for a straight line in networking graphics to represent an Ethernet LAN including a simple star and an extended star.

Physical Topologies



6.1.3 Providing Access to Media

Each network environment that packets encounter as they travel from a local host to a remote host can have different characteristics. For example, an Ethernet LAN usually consists of many hosts contending for access on the network medium. The MAC sublayer resolves this. With serial links the access method may only consist of a direct connection between only two devices, usually two routers. Therefore, they do not require the techniques employed by the IEEE 802 MAC sublayer.

Router interfaces encapsulate the packet into the appropriate frame. A suitable media access control method is used to access each link. In any given exchange of network layer packets, there may be numerous data link layers and media transitions.

At each hop along the path, a router performs the following Layer 2 functions:

1. Accepts a frame from a medium
2. De-encapsulates the frame
3. Re-encapsulates the packet into a new frame
4. Forwards the new frame appropriate to the medium of that segment of the physical network

Press play to view the animation. The router in the figure has an Ethernet interface to connect to the LAN and a serial interface to connect to the WAN. As the router processes frames, it will use data link layer services to receive the frame from one medium, de-encapsulate it to the Layer 3 PDU, re-encapsulate the PDU into a new frame, and place the frame on the medium of the next link of the network.

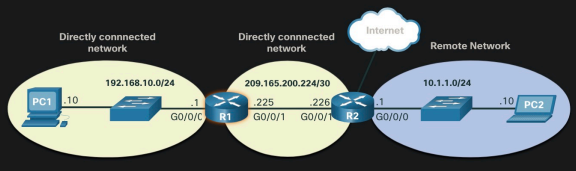
8.5.2 IP Router Routing Table

The routing table of the router contains network route entries listing all the possible known network destinations.

The routing table stores three types of route entries:

- **Directly-connected networks** - These network route entries are active router interfaces. Routers add a directly connected route when an interface is configured with an IP address and is activated. Each router interface is connected to a different network segment. In the figure, the directly-connected networks in the R1 IPv4 routing table would be 192.168.10.0/24 and 209.165.200.224/30.
- **Remote networks** - These network route entries are connected to other routers. Routers learn about remote networks either by being explicitly configured by an administrator or by exchanging route information using a dynamic routing protocol. In the figure, the remote network in the R1 IPv4 routing table would be 10.1.1.0/24.
- **Default route** - Like a host, most routers also include a default route entry, a gateway of last resort. The default route is used when there is no better (longer) match in the IP routing table. In the figure, the R1 IPv4 routing table would most likely include a default route to forward all packets to router R2.

The figure identifies the directly connected and remote networks of router R1.



R1 has two directly connected networks:

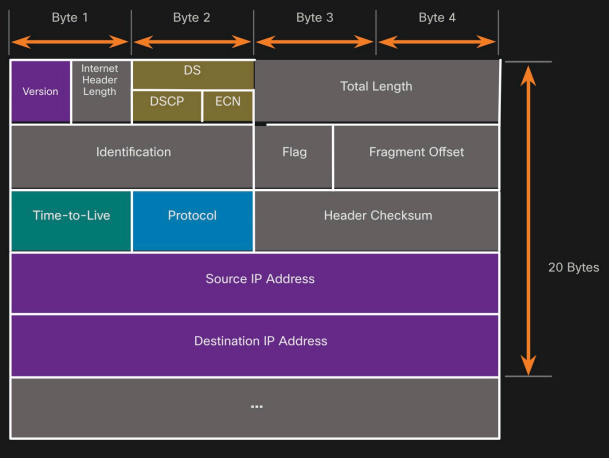
- *192.168.10.0/24*
- *209.165.200.224/30*

R1 also has remote networks (i.e. 10.1.1.0/24 and the internet) that it can learn about.

8.2.2 IPv4 Packet Header Fields

The binary values of each field identify various settings of the IP packet. Protocol header diagrams, which are read left to right, and top down, provide a visual to refer to when discussing protocol fields. The IP protocol header diagram in the figure identifies the fields of a IPv4 packet.

Fields in the IPv4 Packet Header



9.2.2 ARP Functions

When a packet is sent to the data link layer to be encapsulated into an Ethernet frame, the device refers to a table in its memory to find the MAC address that is mapped to the IPv4 address. This table is stored temporarily in RAM memory and called the ARP table or the ARP cache.

The sending device will search its ARP table for a destination IPv4 address and a corresponding MAC address.

- If the packet's destination IPv4 address is on the same network as the source IPv4 address, the device will search the ARP table for the destination IPv4 address.
- If the destination IPv4 address is on a different network than the source IPv4 address, the device will search the ARP table for the IPv4 address of the default gateway.

In both cases, the search is for an IPv4 address and a corresponding MAC address for the device.

Each entry, or row, of the ARP table binds an IPv4 address with a MAC address. We call the relationship between the two values a map. This simply means that you can locate an IPv4 address in the table and discover the corresponding MAC address. The ARP table temporarily saves (caches) the mapping for the devices on the LAN.

If the device locates the IPv4 address, its corresponding MAC address is used as the destination MAC address in the frame. If there is no entry is found, then the device sends an ARP request.

Click Play in the figure to see an animation of the ARP function.

7.3.2 Switch Learning and Forwarding

The switch dynamically builds the MAC address table by examining the source MAC address of the frames received on a port. The switch forwards frames by searching for a match between the destination MAC address in the frame and an entry in the MAC address table.

Click the Learn and Forward buttons for an illustration and explanation of this process.

Learn | Forward

Find the Destination MAC Address

If the destination MAC address is a unicast address, the switch will look for a match between the destination MAC address of the frame and an entry in its MAC address table. If the destination MAC address is in the table, it will forward the frame out the specified port. If the destination MAC address is not in the table, the switch will forward the frame out all ports except the incoming port. This is called an unknown unicast.

As shown in the figure, the switch does not have the destination MAC address in its table for PC-D, so it sends the frame out all ports except port 1.

Note: If the destination MAC address is a broadcast or a multicast, the frame is also flooded out all ports except the incoming port.

10.3.1 Default Gateway on a Host

If your local network has only one router, it will be the gateway router and all hosts and switches on your network must be configured with this information. If your local network has multiple routers, you must select one of them to be the default gateway router. This topic explains how to configure the default gateway on hosts and switches.

For an end device to communicate over the network, it must be configured with the correct IP address information, including the default gateway address. The default gateway is only used when the host wants to send a packet to a device on another network. The default gateway address is generally the router interface address attached to the local network of the host. The IP address of the host device and the router interface address must be in the same network.

For example, assume an IPv4 network topology consisting of a router interconnecting two separate LANs. G0/0/0 is connected to network 192.168.10.0, while G0/0/1 is connected to network 192.168.11.0. Each host device is configured with the appropriate default gateway address.

In this example, if PC1 sends a packet to PC2, then the default gateway is not used. Instead, PC1 addresses the packet with the IPv4 address of PC2 and forwards the packet directly to PC2 through the switch.